Foscount version 0.1.0

READ THIS FIRST

Install and first count

Foscount counts cells automatically in microscope photos of a Bürker counting chamber, for unstained samples and for trypan blue live/dead counts. This page gets you from download to your first result in about five minutes.

1. Install (Windows 10 / 11, 64-bit)

1. Open the download page: **https://github.com/Baboe/burkercell/releases/latest**
2. Under **Assets**, download **Foscount-Setup-x.y.z.exe** (the installer). No administrator rights are needed.
3. Double-click the file. Windows will probably show a blue **"Windows protected your PC"** window, because the app is not code-signed yet. This is expected: click **More info**, then **Run anyway**.
4. Follow the installer. Tick "Create a desktop shortcut" if you want one, then start **Foscount**.

No installation possible? Download **Foscount-portable-x.y.z.exe** instead. It is a single file that runs from any folder (for example a USB stick) but starts more slowly. To remove the installed version: Windows Settings → Apps → Foscount → Uninstall.

2. Your first count

1. **Open images** (top right) or drag photos onto the window. Foscount finds the counting grid and selects the complete squares.
2. In the right-hand panel, choose the **Mode** (unstained, or trypan blue live/dead) and type the **Dilution factor** (1 = undiluted, 2 = mixed 1:1 with trypan blue).
3. **Check the square size** (see the box below).
4. Look at the image. Add, remove or re-classify cells by clicking (*Edit cells* tool). Include or exclude squares with *Select squares*.
5. Read the concentration (cells/mL) and, in trypan mode, the viability at the top of the right-hand panel. Export with **Export results** (Excel or CSV).

Important: set the square size.
Foscount can see the squares but cannot know how big they really are. Under **Chamber → Counting square side**, enter the side of the squares in your photo: **0.2 mm** for the large Bürker squares (the default), **0.05 mm** for the small squares. If a photo shows a uniform grid without the narrow double-line strips, the app warns you. A wrong value gives a wrong concentration (0.05 mm instead of 0.2 mm is a factor of 16).

3. No camera on your microscope?

Use the **Manual entry** tab. Type your counts per square, or use the keyboard tally counter: **L** = live/cell, **D** = dead, **N** = next square, **Backspace** = undo. The same calculations and export are used.

4. Many photos at once

File → Batch process folder counts every photo in a folder and writes an Excel/CSV file plus annotated images. Each subfolder is treated as one sample.

5. Good to know

- **Always check the result by eye.** Clumps, debris and out-of-focus cells can be miscounted. Correct them by clicking.
- Count enough cells: at least 100–200 in total for about 10–20 % precision. The app shows the counting error and warns when counts are low.
- Sharp focus on the grid lines and an evenly lit field give the best results.
- Photos taken through the eyepiece (round view, black corners) work: the black surround is ignored.
- Foscount is a research tool and has not been validated for clinical or diagnostic use.

Help and updates

New versions, source code and bug reports: https://github.com/Baboe/burkercell.
When you report a problem, please attach the photo and say which mode and square size you used.